Dual General Position in the Join of a Complete Graph and a Tree: Structure and Linear-Time Optimization

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Abstract

Given an integer $r \geq 1$ and a tree T of order at least three, we consider the optimization problem of determining the dual general-position number of $K_r + T$ and constructing a maximum set. An exhaustive structural split according to whether a selected set meets the designated complete factor $C = V(K_r)$ reduces the problem to two branches. The C-meeting branch is governed by partitions of T into two induced cliques, while the C-avoiding branch is the tree optimization parameter $\beta(T)$, where $\beta(T) = \max\{|X| : X \subseteq V(T),\ \Delta(T[X]) \leq 1, |N_T(x) \setminus X| \leq 1 \text{ for every } x \in X\}$. This gives

$$\mathrm{gp}_d(K_r + T) = \begin{cases} r + 2, & T \in \{P_3, P_4\}, \\ \beta(T), & \text{otherwise.} \end{cases}$$

We characterize every β-feasible set by vertex degrees and selected neighbors. The characterization yields a four-state rooted-tree dynamic program that computes $\beta(T)$ and reconstructs an attaining tree set in $O(|V(T)|)$ time and space. Combined with the explicit exceptional cases, this gives a maximum dual general-position set for $K_r + T$ in time linear in the order of the join. Worked families include stars, once-subdivided stars, and paths. Bounded exhaustive and definition-first checks test the implementation; they support, but do not replace, the proofs.

Keywords: dynamic programming, combinatorial optimization, dual general position, graph join, tree, graph convexity
2020 MSC: 05C12, 05C05, 05C69, 05C76, 05C85

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1. Introduction

A set of vertices of a connected graph is in *dual general position* if it is in general position and its complement is convex. Equivalently, no geodesic contains three selected vertices, and every geodesic between two unselected vertices stays outside the selected set. Tian and Klavžar introduced and developed this variant of the general-position problem and proved the criterion that combines these two conditions [1]. For a recent overview of general-position problems and their variants, see [2].

From an optimization viewpoint, both conditions are global: they are phrased in terms of geodesics and convexity, and direct enumeration examines exponentially many vertex subsets. For the compactly specified mixed join $K_r + T$, the aim here is therefore not only to determine the optimum value, but also to expose local tree constraints that permit a maximum set to be constructed efficiently. This is an algorithmic graph-optimization objective; no external application of the parameter is assumed.

The standard general-position problem has also been studied under graph operations. In particular, Ghorbani et al. [3] determined the standard general-position number for joins and several other operations. Those results concern standard general position and do not determine the dual general-position number considered here.

The behavior of dual general position under graph products was subsequently studied by Dokyeesun, Klavžar, Kuziak, and Tian [4]. In particular, their work posed the problem of determining the dual general position number of lexicographic products with a complete first factor. Jiang's 2026 preprint [5, Theorem 3.2] gives the formula

$$\mathrm{gp}_d(K_m \circ G) = mq_2(G), \qquad m \geq 2,$$

for every nonempty finite simple graph G, without a connectedness assumption on G, where $q_2(G)$ is the largest size of one class in a partition of $V(G)$ into two sets that each induce a clique. Empty classes are allowed and are regarded as inducing the empty clique; set $q_2(G) = 0$ if no such partition exists. The same preprint [5, Theorem 5.1] also treats complete joins whose factors are all nonempty and noncomplete. Its Theorem 5.1 explicitly does not classify joins containing both a complete and a noncomplete factor. Thus the mixed join $K_r + T$, with T a noncomplete tree, lies outside that theorem; it is also different from the lexicographic product $K_m \circ T$.

One subfamily of the mixed-join problem is already known. The fan graph is $F_n = K_1 + P_n$, and Tian, Dokyeesun, and Klavžar [6, arXiv v2] proved, for $n \geq 4$, that

$$\mathrm{gp}_d(F_n) = \left\lfloor \frac{2(n+1)}{3} \right\rfloor = \left\lceil \frac{2n}{3} \right\rceil .$$

Consequently, the fan case is prior work and is used here only as a consistency check. The formula above is the one displayed in arXiv v2 of [6]; no argument below relies on the different rounding convention displayed in an earlier author-hosted version.

This paper studies $K_r + T$ for $r \geq 1$ and for trees T of order at least three. Its contribution is structural and algorithmic. For a tree T, define

$$\beta(T) = \max\Big\{|X| : \ \Delta(T[X]) \leq 1 \text{ and } |N_T(x) \setminus X| \leq 1 \text{ for every } x \in X\Big\}.$$

The main structural argument separates dual general-position sets that meet the designated complete factor $V(K_r)$ from those that avoid it. The first branch is controlled by a two-clique partition of the tree, while the second is exactly the optimization problem defining $\beta(T)$. This yields

$$\mathrm{gp}_d(K_r + T) = \begin{cases} r+2, & T \in \{P_3, P_4\}, \\ \beta(T), & \text{otherwise.} \end{cases}$$

The main contributions are as follows. First, Sections 3–4 prove the exhaustive two-branch classification, the displayed optimum formula, and an exact local characterization of β-feasible sets. Second, Section 5 turns that local characterization into a four-state rooted-tree dynamic program. It computes $\beta(T)$ and reconstructs an attaining tree set in linear time and space; combined with the explicit P_3 and P_4 cases, this constructs a maximum dual general-position set for the mixed join. Third, Section 6 gives worked families, while Section 7 records bounded exhaustive and definition-first verification routes and their different logical roles. These computations test the implementation and are supporting evidence rather than proof. The comparison with the closest product, join, and fan results is given above; no claim about graph classes outside the stated scope is needed below.

2. Preliminaries

All graphs in this paper are finite and simple. For a graph G, its vertex and edge sets are denoted by $V(G)$ and $E(G)$. If $X \subseteq V(G)$, then $G[X]$ is the subgraph induced by X, and $G - X$ abbreviates $G[V(G) \setminus X]$. The open neighborhood of a vertex x is $N_G(x)$, and $\Delta(G)$ denotes the maximum degree of G; by convention, the empty graph has maximum degree zero. A complete graph of order r is denoted by K_r, and P_n denotes the path of order n.

Let G be connected. The distance $d_G(u,v)$ is the minimum length of a u,v-path, where length means the number of edges. A path of length $d_G(u,v)$ is a *u,v-geodesic*. There may be more than one geodesic between the same two vertices. The interval between u and v is

$$I_G(u,v) = \{w \in V(G) : w \text{ lies on some } u,v\text{-geodesic}\}.$$

A set $Y \subseteq V(G)$ is *convex* if $I_G(u,v) \subseteq Y$ for every $u,v \in Y$; equivalently, every geodesic between two vertices of Y lies entirely in $G[Y]$.

For $X \subseteq V(G)$, a pair $u,v \in V(G)$ is called *X-positionable* if

$$I_G(u,v) \cap X \subseteq \{u,v\}.$$

Thus no internal vertex of any u,v-geodesic belongs to X. The set X is a *general-position set* if every pair of vertices in X is X-positionable. Equivalently, no three distinct vertices of X lie on a common geodesic.

A set $X \subseteq V(G)$ is a *dual general-position set* if every pair whose two endpoints both lie in X, or both lie in $V(G) \setminus X$, is X-positionable. The first family of pairs says exactly that X is a general-position set. By Tian and Klavžar [1, Theorem 3.1], the second family is equivalently expressed by convexity of the complement. We will therefore use the following criterion throughout:

$$X \text{ is dual general position in } G \quad \Longleftrightarrow \quad \begin{array}{l} X \text{ is general position} \\ \text{and } G - X \text{ is convex.} \end{array}$$

The *dual general position number* is

$$\mathrm{gp}_d(G) = \max\{|X| : X \subseteq V(G) \text{ is a dual general-position set}\}.$$

For vertex-disjoint graphs G and F, their *complete join* $G + F$ is obtained from their disjoint union by adding every edge with one endpoint in $V(G)$ and the other in $V(F)$. In the rest of the note, $r \geq 1$, T is a tree of order at least three, and

$$C = V(K_r), \qquad H = K_r + T.$$

Every vertex of C is adjacent to every other vertex of H. Two vertices on the tree side are adjacent in H exactly when they are adjacent in T. Because such a tree is noncomplete, H has diameter two. The order assumption excludes $T \in \{K_1, K_2\}$, for which $K_r + T$ is complete and the diameter-two analysis below does not apply.

We use two parameters on the tree side. First, define

$$q_2(T) = \max\{|A| : V(T) = A \,\dot\cup\, B,\ T[A] \text{ and } T[B] \text{ are complete}\},$$

and put $q_2(T) = 0$ if no such partition exists. Empty classes are allowed and induce the empty clique. The two classes may be interchanged, so $q_2(T)$ records the largest possible class in a partition of the tree into two induced cliques. Second, define

$$\beta(T) = \max\Big\{|X| : X \subseteq V(T),\ \Delta(T[X]) \leq 1, \\ |N_T(x) \setminus X| \leq 1 \text{ for every } x \in X\Big\}.$$

The first constraint in the definition of $\beta(T)$ says that a selected vertex has at most one selected tree neighbor. The second says that it also has at most one unselected tree neighbor. These are conditions on the tree T, not on the additional edges of the join H.

3. The two-branch structural theorem

We first classify dual general-position sets that contain at least one vertex of the designated complete factor C. This is the *C-meeting branch.* Although every vertex of C is universal, some trees, such as stars, contribute an additional universal vertex on the tree side; the terminology refers only to the fixed factor C.

3.1. Sets meeting the designated complete factor

Lemma 3.1. *Let $X \subseteq V(H)$ satisfy $X \cap C \neq \varnothing$, and put*

$$S = X \cap V(T).$$

Then X is a dual general-position set of H if and only if both $T[S]$ and $T[V(T) \setminus S]$ are complete.

Proof. Choose a vertex $c \in X \cap C$. Suppose first that X is a dual general-position set. If two vertices $u, v \in S$ were nonadjacent in T, then they would also be nonadjacent in H. Since c is adjacent to every vertex of H, the path u, c, v would be a u, v-geodesic of length two containing three vertices of X. This would contradict the general-position property of X. Hence $T[S]$ is complete.

Now suppose that two vertices $u, v \in V(T) \setminus S$ were nonadjacent in T. Again u, c, v would be a u, v-geodesic of length two. Its endpoints would lie in $H - X$, whereas its internal vertex c would lie in X. Consequently $H - X$ would not be convex, a contradiction. Thus $T[V(T) \setminus S]$ is complete.

Conversely, suppose that both induced tree subgraphs are complete. The set X is then a clique: its vertices in C are pairwise adjacent, its tree part S induces a clique, and every vertex of C is adjacent to every vertex of T. Therefore X is in general position. The complement $H - X$ is also a clique, because $C \setminus X$ is a clique, $T[V(T) \setminus S]$ is a clique, and all edges between these two parts are present. Every clique is convex, so the criterion from Section 2 shows that X is a dual general-position set. This proves the equivalence. $\square$

The converse also covers the allowed boundary case $X \cap C = C$, equivalently $C \setminus X = \varnothing$. In the present scope, T is noncomplete, so the two tree parts in Lemma 3.1 cannot be empty in a feasible C-meeting set: otherwise the other tree part would induce all of T.

Corollary 3.2. *A dual general-position set meeting C exists if and only if $q_2(T) > 0$. When this branch exists, its maximum cardinality is*

$$r + q_2(T).$$

Proof. By Lemma 3.1, the tree part S of such a set and its complement form a partition of $V(T)$ into two induced cliques. Thus the branch can exist only if $q_2(T) > 0$. Conversely, if $V(T) = A \,\dot{\cup}\, B$ is a partition into two induced cliques, then Lemma 3.1 shows that $C \cup A$ is a dual general-position set meeting C. This proves the existence equivalence.

For every set in this branch, $|X \cap C| \le r$ and $|S| \le q_2(T)$, so $|X| \le r + q_2(T)$. Choose a two-clique partition whose larger designated class has size $q_2(T)$ and include that class together with all r vertices of C. Lemma 3.1 gives a dual general-position set of size $r + q_2(T)$, proving that the bound is attained. $\square$

3.2. Sets avoiding the designated complete factor

We next consider sets contained entirely in the tree side. This is the *C-avoiding branch*; the whole designated complete factor belongs to the complement.

Lemma 3.3. *Let $X \subseteq V(T)$, equivalently $X \cap C = \varnothing$. Then X is a dual general-position set of H if and only if*

$$\Delta(T[X]) \le 1$$

and

$$|N_T(x) \setminus X| \le 1 \qquad \textit{for every } x \in X.$$

Proof. Recall that H has diameter two. We first examine the general-position condition. Suppose that some $x \in X$ has two distinct neighbors $u, v \in X$ in the tree. Since a tree contains no triangle, u and v are nonadjacent. Hence u, x, v is a u, v-geodesic of length two in H, and all three of its vertices belong to X. Thus X is not in general position.

Conversely, suppose that X is not in general position. Then some geodesic contains three distinct vertices of X. Because H has diameter two, these three vertices must occur as u, x, v on a length-two u, v-geodesic, with x as its internal vertex. All three lie in $V(T)$, and adjacency between two tree-side vertices in H is the same as adjacency in T. Therefore u and v are two selected tree neighbors of x. We have proved that X is in general position if and only if no selected vertex has two selected tree neighbors, which is exactly the condition $\Delta(T[X]) \le 1$.

It remains to characterize convexity of the complement. Put $Y = V(H) \setminus X$. Since $X \subseteq V(T)$, the entire clique C lies in Y. Any pair of vertices of Y with at least one endpoint in C is adjacent. Likewise, two tree-side vertices of Y that are adjacent in T are adjacent in H. Such adjacent pairs have geodesics of length one and cannot witness a failure of convexity. Consequently Y can fail to be convex only when two nonadjacent unselected tree vertices have a geodesic of length two whose internal vertex lies in X.

Because every vertex of X is on the tree side, such a geodesic exists exactly when some $x \in X$ has two distinct neighbors $u, v \in V(T) \setminus X$. Indeed, if such u and v exist, then they are nonadjacent because T has no triangle, so u, x, v is a length-two geodesic with endpoints in Y and internal vertex outside Y. Conversely, every length-two geodesic witnessing nonconvexity has a selected internal vertex adjacent in T to its two unselected endpoints. It follows that Y is convex if and only if $|N_T(x) \setminus X| \leq 1$ for every $x \in X$.

Combining the general-position and convexity equivalences with the criterion from Section 2 proves the lemma. □

Corollary 3.4. *The maximum cardinality of a dual general-position set that avoids C is $\beta(T)$.*

Proof. Lemma 3.3 says that the dual general-position sets in this branch are exactly the subsets of $V(T)$ satisfying the two constraints in the definition of $\beta(T)$. Taking the maximum of their cardinalities gives $\beta(T)$. □

3.3. Combining the two branches

Theorem 3.5. *For every $r \geq 1$ and every tree T of order at least three,*

$$\mathrm{gp}_d(K_r + T) = \begin{cases} \beta(T), & q_2(T) = 0, \\ \max\{\beta(T), r + q_2(T)\}, & q_2(T) > 0. \end{cases}$$

Proof. Every vertex set $X \subseteq V(H)$ either meets C or avoids C, and the two cases are disjoint. If $q_2(T) = 0$, Corollary 3.2 says that the C-meeting branch is empty, while Corollary 3.4 gives an attainable maximum of $\beta(T)$ in the C-avoiding branch. If $q_2(T) > 0$, both branches exist: their attainable maxima are respectively $r + q_2(T)$ by Corollary 3.2 and $\beta(T)$ by Corollary 3.4. Taking the larger of these two values proves the formula. □

For trees, the occurrence of the second line of Theorem 3.5 has a particularly short classification.

Lemma 3.6. *If T is a tree of order at least three, then*

$$q_2(T) = \begin{cases} 2, & T \in \{P_3, P_4\}, \\ 0, & \textit{otherwise}. \end{cases}$$

Proof. A tree contains no triangle. Hence every clique in T has at most two vertices. If $V(T)$ can be partitioned into two sets that each induce a clique, then $|V(T)| \leq 4$.

There is only one tree of order three up to isomorphism, namely P_3. Its two adjacent vertices and its remaining vertex form a two-clique partition, so $q_2(P_3) = 2$.

At order four, both classes in a two-clique partition must have size two and therefore must be edges. Thus the two classes form a perfect matching. The two trees of order four are P_4 and $K_{1,3}$. The path P_4 has a perfect matching, obtained by taking its first and last edges, whereas two disjoint edges cannot occur in $K_{1,3}$ because every edge contains its center. Consequently $q_2(P_4) = 2$ and $q_2(K_{1,3}) = 0$. Trees of larger order cannot admit such a partition, proving the lemma. $\square$

We also need the two corresponding values of β.

Lemma 3.7. *We have $\beta(P_3) = 2$ and $\beta(P_4) = 3$.*

Proof. Let the vertices of P_3 occur in the order v_1, v_2, v_3. The set $\{v_1, v_3\}$ is β-feasible: it is independent, and each of its vertices has only the neighbor v_2 outside the set. Hence $\beta(P_3) \geq 2$. The full vertex set is not feasible because its induced subgraph has maximum degree two. Since it is the only set of size three, $\beta(P_3) = 2$.

Now let v_1, v_2, v_3, v_4 be the vertices of P_4 in path order. The set $\{v_1, v_2, v_4\}$ is β-feasible: its induced graph consists of the edge v_1v_2 and the isolated vertex v_4, while v_2 and v_4 each have exactly one neighbor outside the set and v_1 has none. Thus $\beta(P_4) \geq 3$. The full vertex set is not feasible because its induced subgraph has maximum degree two, so $\beta(P_4) \leq 3$. Therefore $\beta(P_4) = 3$. $\square$

Combining Theorem 3.5 with Lemmas 3.6 and 3.7 yields the announced form.

Corollary 3.8. *For every $r \geq 1$ and every tree T of order at least three,*

$$\mathrm{gp}_d(K_r + T) = \begin{cases} r + 2, & T \in \{P_3, P_4\}, \\ \beta(T), & \textit{otherwise.} \end{cases}$$

Proof. If $T \notin \{P_3, P_4\}$, Lemma 3.6 gives $q_2(T) = 0$, so the first line of Theorem 3.5 yields $\mathrm{gp}_d(K_r + T) = \beta(T)$. For $T \in \{P_3, P_4\}$, the second line gives

$$\mathrm{gp}_d(K_r + T) = \max\{\beta(T), r + 2\}.$$

Here $\beta(P_3) = 2$ and $\beta(P_4) = 3$ by Lemma 3.7, while $r + 2 \geq 3$. Thus the displayed maximum is $r + 2$ in both cases. □

4. The tree parameter $\boldsymbol{\beta(T)}$

Call a set $X \subseteq V(T)$ *β-feasible* if it satisfies the two constraints in the definition of $\beta(T)$. Those constraints admit the following equivalent description using only the degrees in the original tree and the labels of the immediate neighbors.

Proposition 4.1. *A set $X \subseteq V(T)$ is β-feasible if and only if both of the following conditions hold:*

1. *no vertex of degree at least three in T belongs to X; and*
2. *every vertex $x \in X$ of degree two in T has exactly one neighbor in X.*

No additional constraint is imposed at a selected vertex of degree zero or one; its label can still affect the constraint at a neighboring selected degree-two vertex. In particular, the statement includes these boundary degrees even though a degree-zero vertex does not occur under the standing assumption that T has order at least three.

Proof. For each selected vertex $x \in X$, put

$$s_X(x) = |N_T(x) \cap X|, \qquad u_X(x) = |N_T(x) \setminus X|.$$

Then

$$s_X(x) + u_X(x) = \deg_T(x).$$

The condition $\Delta(T[X]) \leq 1$ says exactly that $s_X(x) \leq 1$ for every $x \in X$, while the second constraint in the definition of $\beta(T)$ says that $u_X(x) \leq 1$. Hence a selected vertex can have degree at most two in T. If its degree is two, the two nonnegative integers $s_X(x)$ and $u_X(x)$ sum to two and are both at most one, so they must both equal one. This proves the necessity of conditions 1 and 2.

Conversely, assume conditions 1 and 2. If $x \in X$ has degree two, then it has one selected and one unselected neighbor, so $s_X(x) = u_X(x) = 1$. If

$x \in X$ has degree zero or one, then both neighbor counts are automatically at most one. Condition 1 excludes every other degree for a selected vertex. Thus $s_X(x) \leq 1$ and $u_X(x) \leq 1$ for every $x \in X$, which are precisely the two defining constraints of a β-feasible set. $\square$

This equivalence concerns every feasible set, not only the maximum ones. One immediate consequence gives a useful general lower bound.

Corollary 4.2. *If $L(T)$ is the set of leaves of a tree T, then $L(T)$ is β-feasible. Consequently,*

$$\beta(T) \geq |L(T)|.$$

Proof. Every vertex in $L(T)$ has degree one. Proposition 4.1 therefore imposes no additional condition on the set consisting of all leaves. The inequality follows from the definition of $\beta(T)$. $\square$

The first constraint $\Delta(T[X]) \leq 1$ alone defines a weaker maximum-degree-one problem. Every β-feasible set satisfies that constraint, but the converse is false. In $K_{1,3}$, for example, the center together with one leaf induces a single edge and satisfies the weaker constraint. It is not β-feasible, because the selected center has two unselected neighbors. Thus a result or algorithm for the weaker problem cannot simply be substituted for $\beta(T)$ without enforcing the additional outside-neighbor constraint.

The two optimization problems can also have different optimum values. Let B be the tree with root z, children a, b, and two leaves adjacent to each of a and b. Proposition 4.1 forces the degree-three vertices a, b to be unselected. It then also forces z to be unselected, because a selected degree-two vertex must have exactly one selected neighbor. The four leaves form a feasible set, so $\beta(B) = 4$. In contrast, the set consisting of z and the four leaves is independent and therefore has induced maximum degree zero. Hence the optimum in the weaker problem is at least five. This strict separation shows that optimizing only the induced maximum-degree-one constraint can return an incorrect value for β. The bounded literature audit did not identify an established name for this stricter parameter; this paper makes no claim that the parameter itself is new.

5. Linear-time computation and reconstruction

We now give a dynamic program that computes $\beta(T)$ and reconstructs an attaining set. Root T at an arbitrary vertex ρ. For a vertex v, let T_v be the subtree induced by v and all its descendants, and let $\mathrm{Ch}(v)$ be the set of children of v. We use the label 1 for a selected vertex and 0 for an unselected vertex.

For a nonroot vertex v and labels $a, b \in \{0,1\}$, define $F_v(a,b)$ as the largest number of selected vertices in T_v among all labelings with the following properties:

- v has label a;
- the parent of v has label b; and
- every selected vertex in T_v satisfies the two constraints defining a β-feasible set, where the specified parent label is used when checking the constraint at v.

For the root, write $F_\rho(a,\perp)$, where $\perp$ records that no parent exists. An impossible state has value $-\infty$.

Fix a state at v and assign a proposed label $t_u \in \{0,1\}$ to every child $u \in \mathrm{Ch}(v)$. Put

$$s = \sum_{u \in \mathrm{Ch}(v)} t_u, \qquad d = |\,\mathrm{Ch}(v)| + \mathbf{1}_{\{v \neq \rho\}},$$

and define

$$p = \begin{cases} 0, & v = \rho, \\ b, & v \neq \rho. \end{cases}$$

Thus $d = \deg_T(v)$, while $p+s$ is the number of selected neighbors of v. If $a = 1$, the proposed child labels are admissible exactly when

$$p + s \leq 1 \qquad \text{and} \qquad d - (p+s) \leq 1. \tag{5.1}$$

The first inequality bounds the selected neighbors of v and the second bounds its unselected neighbors. If $a = 0$, there is no constraint at v itself, because the definition of β imposes the two neighbor bounds only on selected vertices.

The recurrence is therefore

$$F_v(a,b) = a + \max_{(t_u : u \in \mathrm{Ch}(v))} \sum_{u \in \mathrm{Ch}(v)} F_u(t_u, a), \tag{5.2}$$

where all child-label vectors are allowed when $a = 0$, and only the vectors satisfying (5.1) are allowed when $a = 1$. For the root, the same formula is used with $b = \perp$ and $p = 0$. If the admissible family is empty or a proposed vector uses an impossible child state, its contribution is $-\infty$. When v is a leaf, the empty child-label vector is the unique assignment and its empty sum is zero. The answer is

$$\beta(T) = \max\{F_\rho(0, \perp), F_\rho(1, \perp)\}. \tag{5.3}$$

Theorem 5.1. *The recurrence* (5.2) *computes the exact values of all states, and* (5.3) *equals* $\beta(T)$*. If one maximizing child-label vector is stored for every finite state, a maximum* β*-feasible set can be reconstructed.*

Proof. We prove the state claim by induction on the height of v in the rooted tree. If v is a leaf, it has no child subtrees. When v is selected, (5.1) checks exactly the label of its parent, if one exists; when v is unselected, it has no constraint of its own. Thus (5.2) lists exactly the feasible labelings of the one-vertex subtree and counts its selected vertex by the term a.

Now assume that the claim holds for every child of v. Fix the label a of v and, if v is not the root, the parent label b. Once the child labels are fixed, different child subtrees have no edges between them. Hence no constraint can couple two different child subtrees except through the label of v, which is already fixed in the state. By the induction hypothesis, $F_u(t_u, a)$ is the exact largest contribution from T_u under its two boundary labels. Therefore the best total contribution from the child subtrees is the sum appearing in (5.2).

It remains only to check the constraint at v. If $a = 0$, no such constraint exists, so every combination of feasible child states is allowed. If $a = 1$, then $p + s$ and $d - (p + s)$ are exactly the selected- and unselected-neighbor counts of v. Hence (5.1) accepts precisely the combinations for which v satisfies the two β constraints. The recurrence thus neither omits a feasible labeling nor includes an infeasible one, and the additional term a counts v exactly once. This proves the state claim by induction.

At the root, every labeling of the whole tree has exactly one of the two root labels, so taking the maximum in (5.3) gives precisely $\beta(T)$. To reconstruct an attaining set, choose a maximizing root label. At each visited state, add v

to the set if $a = 1$, read its stored maximizing child-label vector, and continue to each child u in state (t_u, a). These boundary labels are compatible by construction, and the preceding induction shows that every stored child state attains its recorded value. The resulting set is therefore β-feasible and has cardinality $\beta(T)$. □

Proposition 5.2. *The value computation and the reconstruction can both be implemented in $O(|V(T)|)$ time using $O(|V(T)|)$ space.*

Proof. In a state with $a = 0$, the child labels are independent, so for each child u we choose the larger of $F_u(0,0)$ and $F_u(1,0)$. The work is therefore proportional to the number of children. In a state with $a = 1$, (5.1) is impossible when $d \geq 3$, because the two bounded neighbor counts sum to d. Every remaining selected state has at most two children, so at most four child-label vectors need to be examined. There are only constantly many states per vertex. Summing the child counts over all vertices gives $|V(T)| - 1$, and hence the bottom-up computation takes linear time.

Storing a constant number of state values per vertex and one chosen label for each relevant parent–child state uses linear space: summed over all constant-many states, the stored child-vector entries are linear in the total number of tree edges. The top-down reconstruction visits every vertex once and therefore also takes linear time. □

For completeness, the dynamic program also yields a constructive algorithm for the original mixed-join problem. If $T \notin \{P_3, P_4\}$, the reconstructed β-feasible set, viewed as a subset of the tree factor, is a maximum dual general-position set by Corollary 3.8. If T is P_3 or P_4, take all vertices of K_r together with a two-vertex clique class in the corresponding two-clique partition of T. Detecting the exceptional paths takes linear time in the tree order, and listing the complete-factor vertices takes $O(r)$ time. Thus a maximum set for $K_r + T$ is constructed in $O(r + |V(T)|)$ time, which is linear in the order of the join.

6. Known subfamilies and worked examples

The examples below illustrate the local structure of $\beta(T)$ and the two different branches of the main theorem. Their values are derived from the preceding proofs; no computation is being used as proof.

6.1. *Stars and failure of an additive extrapolation*

Let $T = K_{1,k}$ with center z and $k \geq 3$ leaves. Proposition 4.1 forces z to be unselected because $\deg_T(z) = k \geq 3$. On the other hand, the set of all k leaves is β-feasible. It follows that

$$\beta(K_{1,k}) = k.$$

The star $K_{1,k}$ is neither P_3 nor P_4 when $k \geq 3$, so Corollary 3.8 gives

$$\mathrm{gp}_d(K_r + K_{1,k}) = k \qquad (r \geq 1,\ k \geq 3). \tag{6.1}$$

Thus the value is independent of the size of the universal clique. In particular,

$$\mathrm{gp}_d(K_r + K_{1,3}) = 3$$

for every $r \geq 1$. Here $q_2(K_{1,3}) = 0$, so the unsupported extrapolation $r + q_2(T)$ would give 4 when $r = 4$, whereas the correct value is 3. The maximum set in the C-avoiding branch consists of the three leaves.

6.2. *Once-subdivided stars*

Let S_k be obtained from $K_{1,k}$ by subdividing every edge exactly once, where $k \geq 3$. Write z for the center, u_i for the degree-two vertex on the ith arm, and v_i for the corresponding leaf. Again z is forced to be unselected. The set

$$X = \{u_i, v_i : 1 \leq i \leq k\} = V(S_k) \setminus \{z\}$$

is β-feasible: every selected u_i has the selected neighbor v_i and the unselected neighbor z, while each v_i has degree one. Since the only remaining vertex is the forced-unselected center, this set is maximum and

$$\beta(S_k) = 2k.$$

The tree S_k has order $2k + 1 \geq 7$, so $q_2(S_k) = 0$ by Lemma 3.6. Hence

$$\mathrm{gp}_d(K_r + S_k) = 2k \qquad (r \geq 1,\ k \geq 3). \tag{6.2}$$

This family also has a value independent of r, but unlike an ordinary star its maximum set contains both leaves and degree-two vertices.

6.3. Paths and the known fan subfamily

We first determine β on paths directly from Proposition 4.1.

Proposition 6.1. *For every $n \geq 3$,*

$$\beta(P_n) = \left\lceil \frac{2n}{3} \right\rceil.$$

Proof. List the path vertices in order as $v_1, \ldots, v_n$. A β-feasible set cannot contain three consecutive vertices: the middle one would be a selected degree-two vertex with two selected neighbors, contrary to Proposition 4.1. Partitioning the path order into consecutive blocks of three, followed by a remainder of size zero, one, or two, therefore gives

$$|X| \leq \begin{cases} 2q, & n = 3q, \\ 2q+1, & n = 3q+1, \\ 2q+2, & n = 3q+2. \end{cases}$$

These three bounds equal $\lceil 2n/3 \rceil$. They are attained by the binary selection words

$$(110)^q, \qquad (110)^q 1, \qquad (110)^q 11,$$

respectively, where a 1 means that the corresponding path vertex is selected. In each word, every selected internal vertex has exactly one selected neighbor, while the two endpoints have degree one. Proposition 4.1 therefore shows that the indicated sets are β-feasible. $\square$

Corollary 3.8 now yields, for every $r \geq 1$ and $n \geq 3$,

$$\mathrm{gp}_d(K_r + P_n) = \begin{cases} r+2, & n \in \{3,4\}, \\ \left\lceil \dfrac{2n}{3} \right\rceil, & n \geq 5. \end{cases} \tag{6.3}$$

When $r = 1$, the graph $K_1 + P_n$ is the fan F_n. For $n = 4$, the first line of (6.3) gives $3 = \lceil 8/3 \rceil$, and for $n \geq 5$ the second line applies. Thus, for every $n \geq 4$, the present theorem recovers

$$\mathrm{gp}_d(F_n) = \left\lceil \frac{2n}{3} \right\rceil = \left\lfloor \frac{2(n+1)}{3} \right\rfloor.$$

For $n = 3$, however, the first line of (6.3) gives $\mathrm{gp}_d(F_3) = 3$, whereas $\lfloor 2(3+1)/3 \rfloor = 2$. This explains why the known fan formula is stated only for $n \geq 4$ and highlights the exceptional P_3 branch.

For $n \geq 4$, the equality of the ceiling and floor expressions follows immediately by considering n modulo three. This is exactly the formula stated in arXiv v2 of [6], so the fan calculation is a consistency check and not a novelty claim. Formula (6.3) additionally makes clear that for paths of order at least five, the value stays unchanged when the single fan apex is replaced by an arbitrary clique K_r; the exceptional paths P_3 and P_4 have a C-meeting set attaining the value $r + 2$. For P_4 with $r = 1$, the C-avoiding branch also attains the same value 3; this tie does not alter the formula.

7. Computational verification and reproducibility

The proofs in Sections 3–6 do not depend on computation. The calculations reported here test the implementation and search for small counterexamples; finite agreement is supporting evidence and is not a proof of any theorem.

7.1. Two verification routes

The project keeps the following two verification routes separate. They share the audit driver and NetworkX-generated tree instances, but they answer different questions.

First, `src/mixed_join_tree.py` implements the rooted-tree recurrence from Section 5 and reconstructs one maximizing set. It does not import the shortest-path checker. The audit compares its value with a separate exhaustive search over all subsets satisfying the two defining local constraints of $\beta(T)$. The comparison covers every nonisomorphic tree of orders 3 through 12 generated by NetworkX, a total of 985 trees. For each tree, the audit also checks that the reconstructed set has the reported size and satisfies the two local constraints. Finally, it recomputes the value with every possible root to test root invariance. Because both sides of this first comparison use the same two local constraints defining $\beta(T)$, it tests the dynamic-program implementation and reconstruction, not the correctness of the reduction from dual general position to those constraints.

Second, `src/dual_gp_independent.py` provides a definition-first checker that does not use the mixed-join theorem, the β characterization, or the tree DP. It constructs $K_r + T$ explicitly, computes all-pairs distances by fresh breadth-first searches, enumerates vertex subsets, and tests general position together with convexity of the complement. The audit compares the maximum found by this checker with the formula implemented from Corollary 3.8. It covers all 46 nonisomorphic trees of orders 3 through 8 and $r \in \{1, 2, 3, 4\}$, giving 184 comparisons. On the same 184 mixed joins, the checker also verifies directly that the tree-side set reconstructed by the dynamic program is dual general position. This latter check concerns feasibility; on P_3 and P_4, a C-meeting set may be larger than the reconstructed C-avoiding set.

The archived report `results/mixed_join_dp_audit.json` records:

Check	Scope	Number checked	Failures
DP value versus exhaustive subset search	all nonisomorphic trees, orders 3–12	985	0
root-invariance comparison	every possible root of the same 985 trees	11,003	0
reconstructed maximum set	one reconstruction for each of the 985 trees	985	0
mixed-join formula versus shortest-path definition	all nonisomorphic trees, orders 3–8, and $r = 1, 2, 3, 4$	184	0
reconstructed tree-side set versus shortest-path definition	the same 184 mixed joins	184	0

The earlier target-selection screen in `results/extension_feasibility_audit.json` contains 92 overlapping small mixed-join comparisons and the explicitly retained $K_{1,3}$ data. Those historical checks are not added to the table totals, because doing so would double-count instances from the larger audit.

OpenAI Codex assisted in drafting and editing the project source code and tests. The author specified the mathematical constraints and audit questions, reviewed the resulting source, executed the reported checks, and compared the dynamic-program output with the two verification routes described above. The public source and fixed archive permit inspection and rerunning of this work; the author remains responsible for the code, reported outputs, and their interpretation.

7.2. Environment and commands

The archived matrix was produced under CPython 3.13.5 and NetworkX 3.6.1. A fresh rerun on 29 August 2026 used Windows 11 build 26100, CPython 3.13.5, NetworkX 3.6.1, and pytest 9.1.1. From the repository root, the relevant PowerShell commands are

```
.\.venv\Scripts\python.exe -m pip install ‘
  -r .\requirements-lock.txt
.\.venv\Scripts\python.exe -m pytest -q .\tests
.\.venv\Scripts\python.exe ‘
  .\experiments\audit_mixed_join_dp.py ‘
  --output .\results\mixed_join_dp_audit.json
```

The fresh test run returned `30 passed`. In addition to the bounded matrices, the tests cover named values, invalid inputs, reconstruction feasibility, and a path of order 2,500 to ensure that reconstruction does not rely on Python recursion. The fresh audit rerun reproduced all five zero-failure rows in the table above.

For exact artifact identification, the SHA-256 hashes recorded on 29 August 2026 are:

Artifact	SHA-256
`src/mixed_join_tree.py`	`D3DDAB46510B217BF9C442CC57302953` `C4D0C29F208DD7ADE06381F2D90A8B9D`
`src/dual_gp_independent.py`	`D7BA1C520DC6991E78CB8BAE609A598E` `6BBF950DA03C1EAC9BA4D30D1EFFF231`

`experiments/`
`audit_extension_candidates`
`.py`
	`A463CE202995000F324BDB7F90D2821B`
	`1B97AB38911E3E8941DFFC381F8393F6`

`experiments/` `audit_mixed_join_dp.py`	`AE55D305A2893B987E348F64273A7F3C` `D49738E202BBCD5ACCE48F7161366DCA`
`results/` `mixed_join_dp_audit.json`	`1FBCD77D2457F5EED9CC1ED518E47D09` `7145EF0DF36648A3F21BA36D9AC1B7F9`

Artifact	SHA-256
`requirements-lock.txt`	`4811AEA9E5C13E192FB5865D7095ECBA` `19A33887ACC1D567AB5368E1D31DDE5E`
`mixed_join_v6_reproducibility.zip`	`0E91BAAC07EFA121784CA94355C93F30` `4A7AF8FF89AB480E952E9C62DC316A33`

The locked requirements file records every package installed in the archived environment. The archive named in the table contains the source, tests, machine-readable reports, proof note, locked environment, and reproduction instructions. It is publicly available in the repository https://github.com/Star5Dust/dual-general-position-mixed-joins as `artifacts/mixed_join_v6_reproducibility.zip`. It was also extracted into a clean directory, where the test suite passed and the main JSON report was reproduced byte for byte. None of these finite checks establishes correctness beyond the stated test ranges.

8. Conclusion and limitations

The main outcome is an exact and constructive solution of the dual general-position optimization problem on $K_r + T$. For $r \geq 1$ and a tree T of order at least three, feasible sets split into two exhaustive classes. A set meeting the designated complete factor exists precisely when the tree can be partitioned into two induced cliques, and the maximum in that branch is $r + q_2(T)$. A set avoiding the designated complete factor is dual general position precisely when it satisfies the two local constraints defining $\beta(T)$, and the maximum in that branch is $\beta(T)$. Since a tree in the present scope has a two-clique partition only when it is P_3 or P_4, the combined formula is

$$\mathrm{gp}_d(K_r + T) = \begin{cases} r + 2, & T \in \{P_3, P_4\}, \\ \beta(T), & \text{otherwise.} \end{cases}$$

The local characterization of β excludes selected vertices of degree at least three and requires every selected degree-two vertex to have exactly one selected neighbor. The four-state rooted-tree dynamic program turns this characterization into a linear-time, linear-space computation of $\beta(T)$ with reconstruction. Combining that output with the explicit P_3 and P_4 branch constructs a maximum dual general-position set for the mixed join in $O(r + |V(T)|)$ time. Stars, once-subdivided stars, and paths illustrate the resulting structure. For $r = 1$, the path calculation recovers the formula stated in arXiv v2 of [6]; that subfamily is prior work and is not presented as a new result.

Several limitations are essential. Jiang's closest general result [5] was available as preprint v1.0.1 at the literature cutoff of 2 September 2026. The bounded computations in Section 7 support the implementation but do not prove the theorem. The comparison with the known fan formula uses the display in arXiv v2 of [6], which has the rounding convention consistent with formula (6.3).

Finally, this paper treats only $K_r + T$ with $r \geq 1$ and $|V(T)| \geq 3$. It does not classify joins with an arbitrary first factor, general mixed complete joins with several factors, or lexicographic products $P_n \circ T$. Those directions are outside the present scope, and no claim about their resolution or novelty is made here. The contribution is an algorithmic result for this graph-optimization family; it is not presented as a real-world application of dual general position.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Yi Yuteng: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The code, tests, locked Python environment, proof note, and machine-readable audit outputs supporting the computational checks are deposited in the public repository at https://github.com/Star5Dust/dual-general-position-mixed-joins. The fixed archive used for the reported checks is `artifacts/mixed_join_v6_reproducibility.zip`; its SHA-256 checksum is recorded in Section 7.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used OpenAI Codex to assist with literature organization, proof and manuscript drafting, code development, computational checking, and editorial revision. After using this tool, the author reviewed and edited the content and source code as needed and takes full responsibility for the content of the published article.

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